Synthetic Ratings, Different Recommendations

Comparing SVD and NMF under controlled category-level injection

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Recommender systems research

Same injected profiles. Different model responses.
Books / Thriller / 500 synthetic users / Top-15

NMF **10.84**

SVD **2.28**

Thriller books per list

01 Research question

How does the same repeated rating pattern change what SVD and NMF recommend to real users?
We measure category exposure: how many target-category items appear in each user's Top-15 list.

02 Three real datasets

BOOKS
53,424 users
9,415 books | 5,815,569 ratings
13 assigned genres; overlapping labels.

MOVIES
943 users
1,682 movies | 100,000 ratings
MovieLens 100K; target: 1990s releases.

MOBILE APPS
700,111 users
10,173 apps | 19,297,019 ratings
48 original categories; target: Shopping.

03 Controlled injection

Every fake user rates the existing catalog using the same rule:

TARGET BOOKS rating 5

ALL OTHER BOOKS rating 1

Thriller example: rating rule (schematic)

5	5	5	1	1	1	1	1

1,606 Thriller + 7,809 non-Thriller = 9,415 ratings

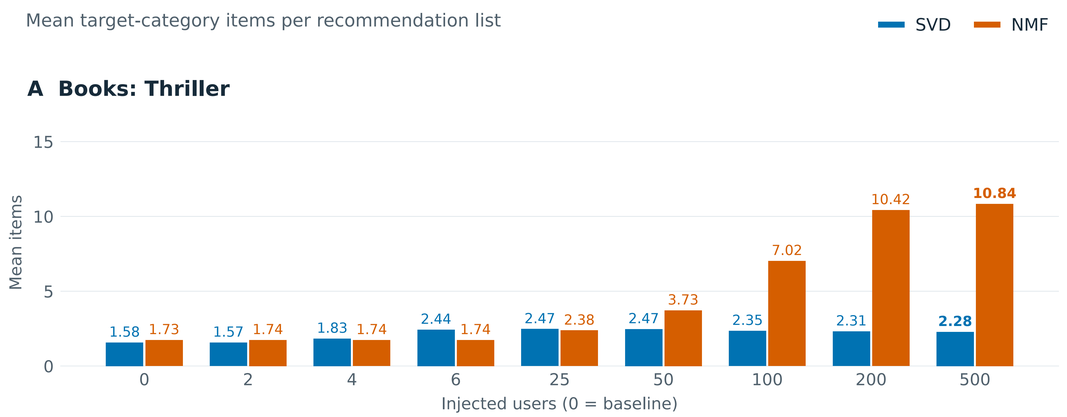
500 fake users add 4,707,500 ratings: 80.9% of the original book rating count.

Original users, ratings and books stay unchanged. Fake users enter training only; recommendations are evaluated for real users.

SHARED TRAINING SETTINGS
8 latent factors | 180 epochs
Biased Surprise implementations; seed 42.

04 What changes in Top-15?

Mean target-category items per recommendation list



All panels count items out of 15. Injection counts are categorical.
Counts and rating budgets differ across datasets.
The movie target already has high baseline exposure.

05 Two factorization models

$$\hat{r}_{ui} = \mu + b_u + b_i + p_u^\top q_i$$

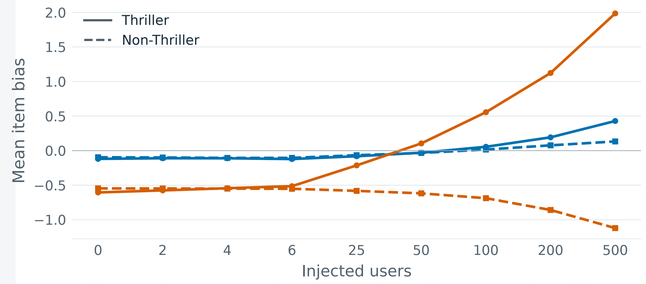
Predicted rating = global mean + user bias + item bias + user-item dot product.

SVD Signed user and item factors; additive gradient updates.

NMF Non-negative factors; multiplicative factor updates. Biases can be negative.

06 A closer look at item bias

Books / Thriller injection / learned b_i



NMF shows a larger target / non-target item-bias separation. This accompanies its stronger Thriller exposure.

Each book has equal weight in this bias panel.

07 What we learn

Exposure changes.
Repeated synthetic ratings reshape the recommendations received by real users.

The model matters.
NMF amplifies Thriller more; SVD has higher Shopping exposure in the displayed mobile range.

The result is diagnostic.
These runs measure exposure and fitted parameters, not held-out accuracy or a validated defense.

DATA: Goodbooks-10K | MovieLens 100K | MobileRec

Book genres: GPT-3.5-assigned labels; exposure is conditional on these labels. Baseline = no synthetic users.

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48 x 36 in | Stage-one results